

Market Garden: Beds, Rotation, the Season

Carpathian OER Commons — Land, Forest, Food Module

Michel Garand | 2026-09-29 | Version v0.5 | CC BY-SA 4.0

Maturity: L0, a marked draft — sized and designed, not yet fully costed, and published so that it can be read and changed in the open. The seed, the tarps, the hoops, the pond, its pump, and the carriage of the bought compost are not priced, and nothing has been grown. Read every figure below at this level and no higher. The module moves to L1 when every line in Section 6 carries a dated cost and its basis.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only. No market, town, buyer, route, or day of distribution is named, and no list of those who eat from the garden enters the repository.

"Market garden" names the method — intensive vegetable growing on permanent beds — and not a sale. The garden serves the hamlet's table first; what leaves the hamlet, once the hamlet is fed, goes through the growers' and eaters' association, a module of its own in Settlement.

*v0.2, 2026-09-28: sized from the ground, about 3,000 m², instead of from demand, and reviewed against Jean-Martin Fortier's *The Market Gardener*, 2014, cited by page. Six plots of sixteen beds; compost from Fortier's own programme; irrigation demand added, and carried by a small pond that the stream refills, decided 2026-09-28.*

v0.3, 2026-09-28: the garden is called ecologically oriented, not organic; no certification is sought for now.

v0.4, 2026-09-29: the establishment compost priced from the compost module; the beds, the compost bays, and the manure heap kept out of the Water Code's protective strip along the stream and around the pond; the connections brought into line with the two streams and with the compost, terra preta, and biochar modules.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

v0.5, 2026-09-29: a note on permits and local rules added at the head of the module; nothing else changed.

1. Purpose

Fresh vegetables for the hamlet's kitchen and table at capacity, grown on permanent beds by a paid grower, on ground fed from the garden's own nutrient stream. What the kitchen cannot use fresh goes to the food storage module; what the hamlet does not need, once it is fed, goes to the association.

Five rules. The hamlet eats first: the region receives only a measured surplus. Two nutrient streams kept apart: the toilet stream feeds trees; only the garden's own stream reaches the beds. The grower is paid: the garden's work is a cost line, never a contribution, and toloka is not the name for work the garden depends on. Ecologically oriented, not certified: the garden describes its practice as ecologically oriented and seeks no certification for now; its produce is not labelled organic, bio, or eco, words Ukrainian law keeps for certified produce. Spring water is for people: the garden is never irrigated from the spring, but from a small pond the stream refills.

Connections. Depends on: the Phase 1 walk of the ground, a soil analysis, and the season measured or taken from the nearest station; the compost module for the garden's own stream, kept apart from the toilet stream, and for the compost bought in; the biochar module for the char charged in that stream; a small pond, always refilled by the stream, as its irrigation source, decided 2026-09-28 — the rainwater and nursery irrigation module keeps the nursery; the energy module for the pump on the essential or seasonal circuit; the water module for washing produce in spring water; the legal form decision, since the garden is a registered capacity of the operator that runs the kitchen. Feeds: the kitchen and dining module; the food storage module, whose cold room the garden needs from the first harvest that is not eaten the same day; the growers' and eaters' association, with its measured surplus only; the compost module, with the residues that leave the beds. The terra preta module does not feed the beds; its soil goes to trees, as decided on 2026-09-28.

Patterns. Grows from: not yet named. Where vegetables were grown in this valley relative to the house, the hayfield, and the slope; how that ground was fed; and who kept the seed, are questions for the elders before the first bed is dug. Feeds: not yet named. Resonates with: 177 Vegetable Garden — ground set aside for vegetables, sunny, fenced, with its tool store beside it; 169 Terraced Slope — permanent beds on the contour; 104 Site Repair — beds on the best ground, the tunnel, cellar, and compost bays on the poorer; 147 Communal Eating — the garden serves one table first; 170 Fruit Trees — where the toilet stream goes, beside the beds and never in them. In tension with: 178 Compost — Alexander joins toilets and kitchen waste into one fertiliser; here the two streams are kept apart, and only the garden's own stream reaches the beds. 172 Garden Growing Wild — permanent beds are formal ground; the tension is held at the edges, and by keeping no earth bare. 4 Agricultural Valleys — the best ground here may be hayfield, and a bed there is taken from winter fodder. 46 Market of Many Shops — the region is reached through an association of growers and eaters, not a market.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — mountain; frost into May and from late September

Condition	Range this design holds for
Frost-free period	frost_free_days, [set] — sources: 60 to 160 days
Altitude	[set]
Ground	garden_area: about 3,000 m ² , fenced, [set]
Slope	under 5 percent for open beds; steeper ground terraced
Aspect	south to south-east; not in a frost pocket
Soil	acidic; limed to the analysis before the first bed
Water	about 50 m ³ a dry week, from a pond the stream refills
Protective strip	beds 25 m from stream and pond, 50 m on slopes
Group size	capacity: 34 people, the example below
Days of demand	use_days_per_year: 120, the example below

The frost-free period is read from the literature and not from the site. A general source gives the Carpathians up to 160 frost-free days, with mountain valleys shorter by a further 30 to 40; a weaker source gives 60 to 100 for the mountain zone. The site's own figure comes from the nearest hydrometeorological station, then from the site's own record.

Fortier's garden, from which the bed system and several figures below are taken, is in southern Quebec, hardiness zone 5, on gravelly loam (pp. 45, 147). He sets the ideal site on a gentle, steady south-facing slope under 5 percent, and never at the bottom of a slope or valley, where frost comes earlier (pp. 22–24). In a mountain valley that points the garden up the slope and away from the valley floor — toward the tension between garden and hayfield named in Section 1.

3. Sizing method

v0.1 sized the garden from the kitchen's demand. This version sizes it from the ground — about 3,000 m² — and then reads the result against the demand. Both directions are kept: the ground sets what can be grown, the demand sets what must be.

What you must measure yourself before using this module:

- **Your season** — the last spring and first autumn frost, from the nearest station for as many years as it holds, then from your own thermometer at bed height.

- **Your soil** — a laboratory analysis of pH, humus, and N, P, K, one sample for each kind of ground you mean to use. Fortier aims for pH 6.5 and raises it with limestone in small doses, measured before each application (p. 58).
- **Your ground** — the area of deep, sunny ground that is not needed as hayfield, its slope and aspect, and the length of bed the contour allows.
- **Your water** — the flow of the stream that refills the pond, in its driest month; never the spring.
- **Your demand** — the kitchen's vegetables per person per day, and the days of demand in a year, residents included.

The bed. Width 0.75 m and paths 0.45 m, 1.2 m from centre to centre: Fortier's 30-inch bed and 18-inch path, which he recommends to anyone growing without a tractor because the tools for market gardening are now made to that width (p. 34). Length 20 m, [set], as the contour allows; Fortier's 100-foot beds suit his site and are, in his words, not a standard — what matters is that every bed is the same length, so that tarps, drip lines, and covers are cut once and fit every bed (pp. 34–35). One bed is 15 m².

The plot. Sixteen beds side by side, with 1.8 m at each end for a harvest cart, after Fortier's plots of sixteen beds with 6 feet at either end (p. 35). One plot is 16×1.2 m by $20 + 2 \times 1.8$ m, 19.2 by 23.6 m, about 453 m².

The rotation. Six plots, [set], after Fortier's ten (pp. 66–69). Each year three plots carry a heavy-feeding group and its compost, and three carry greens and roots; the next year they change places. The three groups, as in his rotation, are the Solanaceae; the Brassicaceae with the Cucurbitaceae, which he combined in the same way; and the alliums with garlic. Peas and beans go with greens and roots and take no compost. Each group comes back to the same plot every sixth year. The tunnel is outside the rotation.

Our example: garden_area, about 3,000 m², [set]; capacity, 34 people; use_days_per_year, 120; vegetables_per_person, 0.44 kg; garden_share, 1.0, [set]; bed_yield, 2 kg per m². Values from the site parameters file, held in the workshop.

Quantity	Calculation	Our example
Plots	$6 \times 453 \text{ m}^2$	about 2,720 m ²
Tunnel	$6 \times 20 \text{ m}$	120 m ²
Compost bays, store, washing	the rest of the ground	about 160 m ²
Beds	$6 \text{ plots} \times 16$	96 beds
Bed area, open ground	$96 \times 15 \text{ m}^2$	1,440 m ²
Output, a season	$\text{bed area} \times \text{bed_yield}$	about 2,880 kg

Quantity	Calculation	Our example
Demand, a year	people × days × kg × share	about 1,795 kg
Beds the demand needs	demand ÷ 2 kg ÷ 15 m ²	about 60 beds
Compost, first year	bed area × 4.3 cm	about 62 m ³
Compost, each year after	half the bed area × 3.0 cm	about 22 m ³
Dolomite, first year	bed area × 0.4 kg, [assumed]	about 576 kg
Water, a dry week	(bed area + tunnel) × 32 mm	about 50 m ³
Fence	the perimeter of 3,000 m ² , as a square	about 220 m

The output against the demand. At the assumed yield, the 96 beds give about 2,880 kg a season against a demand of about 1,795 kg: the demand needs about 60 beds, and the other 36 carry the residents' demand, the crops that do not reach the yield, and the years that go badly. The difference is not a surplus. The association's second circle opens only on a surplus measured through a full season, never on this arithmetic.

The compost. Fortier's programme gives heavy feeders 5 wheelbarrows of compost to a bed of 30 inches by 100 feet, each bed once every two years; his wheelbarrows hold 5 cubic feet (p. 56). That is about 3.0 cm of compost at each application. To build his beds he added about 7 wheelbarrows of a compost mix to each bed, about 4.3 cm (p. 45). Charles Dowding's no-dig figures are higher: 5 cm to start, 3 cm a year after, which would be about 72 m³ and 43 m³ here. Fortier advises beginning growers to buy their compost rather than make it (p. 62). The compost module sizes the garden's own stream against this need: the beds are built from bought rotted manure, all of it, since nothing the site makes is finished before the first beds are dug; after that the site makes between a twenty-fifth and a half of what the beds take, and about 11 to 21 m³ is bought every year while the garden runs at this size and no animals are kept. Built in stages, the sixty beds the demand needs take about 39 m³ in the first year and the other thirty-six about 23 m³ the next.

The water. Fortier gives most crops about 1¼ inches of water a week, about 32 mm (p. 39), and advises a pond, lake, or river once more than an acre is cultivated (p. 26). At 1,560 m² of bed and tunnel that is about 50 m³ in a week without rain. The spring is sized for people — 2.55 m³ a day at capacity in the water module — and is never used for the garden. The rainwater cistern of 4.2 m³ in the source working paper was sized for a 50 m² nursery. Neither carries the garden.

The pond. Decided 2026-09-28: a small pond, always refilled by the stream, carries the garden. 50 m³ a week is about 7 m³ a day, or about 5 L a minute running day and night; a stream that gives at least that in its driest month refills a dry week's water as fast as the garden uses it, and the pond only has to hold a

day or two of watering and the depth the pump's intake needs. Its volume is [set] from the stream's measured flow. Fortier's advice for the pump: set it beside the water, since pumps push well and pull badly; fit a sediment filter so the emitters do not clog; and give the main line a relief valve (p. 39). Because children are on the site, the pond is fenced with a gate and shallow at its edges, as the sanitation working paper fences its own; it is a reservoir, not a swimming pond. The Water Code keeps vegetable growing, fertiliser, and manure stores out of a protective strip 25 m wide along the stream and around a pond under 3 ha, doubled where the bank slopes more than 3° (Section 10): no bed, compost bay, or manure heap stands within it, measured from the stream and from the pond's own edge. The pond and the beds are therefore placed apart, with the main line crossing the strip; whether the pond and its intake may be dug inside the stream's strip is asked in the written enquiry.

The dolomite rate is 400 g per m², from a retailer's guide for loam at pH 4.5 to 5.0; an agronomic table gives 4 to 5.5 t of lime a hectare, as CaO, for loam at the same pH, once in six to eight years. The analysis replaces both.

Read the result against three others. Fortier estimates 30 to 70 shares of a 20-week programme for each half acre of intensive garden (p. 21); 3,000 m² is about three quarters of an acre, or about 44 to 104 shares. Alexander's Vegetable Garden gives one-tenth of an acre for a family of four; 3,000 m² at that rate feeds about 30 people. Our example gives about 1,440 m² of bed for 34 people's 120 days of demand, with a margin. The comparisons say the size is not far out; they do not say it is right.

From beds to a crop plan. Fortier plans the season from the demand backwards: the number of times each single-harvest crop is wanted, the units a week each multiple-harvest crop must give, and his table of yield per 100-foot bed, which he says must be scaled to other bed lengths (pp. 134–141). A 20 m bed is about two thirds of his. The crop plan is the kitchen's menu turned into beds, plot by plot, within the rotation; it is written each winter and is not part of this module.

4. Bill of materials

Quantities for the six-plot example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Soil analysis	3 samples, [set]	none named
Dolomite flour	24 bags of 25 kg	lime to the analysis
Rotted manure	about 62 m ³ , first year	compost; two stages

Item	Quantity	Alternatives in the design
Drip tape, 20 cm emitters	3,840 m, 2 lines a bed	micro-sprinklers
Main line, PE 32 mm	about 120 m, [assumed]	none named
Start connectors	192	none named
Filter and pressure regulator	1 set	none named
Small pond, refilled by the stream	volume [set]	none named
Intake from the stream	[set]	none named
Pump, sediment filter, relief valve	1 set, [set]	none named
Pond fence and gate	[set]	none named
Film tunnel	1, 6 × 20 m, double film	caterpillar tunnels
Welded mesh, 2 m high	9 rolls of 25 m	electric fence
Fence posts, 2.5 m	89, 2.5 m apart, [assumed]	timber from the site
Wicket and double gate	1 each	none named
Row cover, 17 g/m ² , 3.2 m	48 rolls of 10 m, half the beds	none
Hoops for row cover	[set]	wire, PVC, conduit
Insect net	7 rolls of 1.2 × 30 m, 10 beds, [set]	none named
Black tarps for occultation	[set], cut to one plot	none named
Hand tools	cultivator, seeder, 2 hoes, ripper, 2 barrows	broadfork
Compost bays	3 bays, 3 × 3 × 1.2 m, [set]	none named
Tool store	1, metal	timber, from the site

Why drip, not sprinklers. Fortier waters open beds with micro-sprinklers and keeps drip for the tunnels and plastic mulch, because drip lines must be lifted before every hoeing; he also writes that drip uses water far more efficiently (p. 37). Here water is the scarce thing, so the design keeps drip on the beds and a short line of mini-sprinklers for keeping direct-seeded beds moist until they germinate, [set].

Why the tunnel and the covers. Fortier puts his whole spring production under row cover of about 19 g per m², which raises soil temperature by 2 to 3 degrees and protects against frost and wind (pp. 120–121). A short mountain season asks the same. Caterpillar tunnels — movable, of pipe, rope, and old film — are his cheaper complement to a fixed tunnel (pp. 122–123).

Why a 2 m fence. Fortier fences against deer with metal fencing six feet high, and calls electric fencing the cheaper option when starting or renting (pp. 35–36).

The layout, as designed. Six plots on the contour, each of sixteen beds; the tunnel, the store, and the compost bays together on the poorer ground at one side, as near the plots as the ground allows — Fortier puts the tool shed, washing, and cold room at the centre of the gardens, so that no plot is far from them (p. 31); the wicket toward the kitchen and the double gate toward the track for deliveries. Every bed, bay, and heap outside the Water Code's strip along the stream and around the pond; the bays and the heap also at the compost module's distances — at least 50 m from the spring and never upslope of it, 20 m from dwellings and places where children are. Drawn as relationships only; the walk sets the rest.

5. Build

The soil analysis and the liming come first; the fence and the tunnel before the first beds; the beds in the order the compost allows. Fortier built his beds once, digging the paths onto the beds and mounding them about 8 inches, 20 cm, which settle to 10 to 15 cm over two seasons (p. 45).

Step	Trade	Time	If done out of order
Walk, take samples	grower	[from L2]	beds on the wrong ground
Stream measured, pond dug	excavator	[from L2]	beds without water
Soil analysis	laboratory	[from L2]	lime and compost guessed
Lime the bed ground	grower	[from L2]	acid-sensitive crops fail
Fence and gates	fencer	[from L2]	first crops browsed
Tunnel frame and film	fitter	[from L2]	[from L2]
Main line, filter, drip	grower	[from L2]	[from L2]

Step	Trade	Time	If done out of order
Beds shaped, compost laid	grower	[from L2]	[from L2]
Compost bays and store	carpenter	[from L2]	[from L2]

Fortier prepares a 100-foot bed in about 30 minutes once the tarp has done its work (p. 42); the time for a 20 m bed here is [from L2].

6. Cost

Estimates, not quotations: each line is a price listed by a Ukrainian retailer or service platform, seen on 2026-09-28. Place: not a quotation; national listings, and one platform in western Ukraine for the tunnel frame. "Listing" below means that listed price. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

6.1 Ground

Item	UAH	EUR	Basis
Soil analysis, 3 samples	2,989	57	estimated, institute's list
Dolomite flour, 24 bags of 25 kg	6,000	115	estimated, listing
Rotted manure, 62 m ³ , about 55.8 t	39,060	751	estimated, listing
Carriage of the manure	[set]	[set]	not priced
Subtotal, priced lines	48,049	924	

6.2 Irrigation

Item	UAH	EUR	Basis
Drip tape, 4 rolls of 1,000 m	10,172	196	estimated, listing
Main line, PE 32 mm, 120 m	4,440	85	estimated; length [assumed]
Start connectors, 192	1,920	37	estimated, listing
Filter and pressure regulator	2,322	45	estimated, listing
Pond, dug and sealed	[set]	[set]	not priced

Item	UAH	EUR	Basis
Intake, pump, sediment filter	[set]	[set]	not priced
Pond fence and gate	[set]	[set]	not priced
Subtotal, priced lines	18,854	363	

6.3 Tunnel

Item	UAH	EUR	Basis
Film tunnel, 6 × 20 m	100,000	1,923	estimated, listing
Frame assembly	4,500	87	estimated, lower bound of a rate
Subtotal	104,500	2,010	

6.4 Fence

Item	UAH	EUR	Basis
Welded mesh, 2 m, 9 rolls of 25 m	20,797	400	estimated, listing
Posts, 2.5 m, 89	38,715	745	estimated; spacing [assumed]
Wicket, 1.0 × 2.0 m	10,801	208	estimated, listing
Double gate, 3 × 2.0 m	17,135	330	estimated, listing
Installation, about 220 m	39,600	762	estimated, listed rate
Wicket and gate installation	3,600	69	estimated, listed rate
Subtotal	130,648	2,512	

6.5 Covers

Item	UAH	EUR	Basis
Row cover, 48 rolls of 3.2 × 10 m	8,352	161	estimated, listing
Insect net, 7 rolls of 1.2 × 30 m	11,151	214	estimated, listing
Hoops	[set]	[set]	not priced

Item	UAH	EUR	Basis
Black tarps	[set]	[set]	not priced
Subtotal, priced lines	19,503	375	

6.6 Tools

Item	UAH	EUR	Basis
Wheeled hand cultivator	1,500	29	estimated, listing
Precision seeder	2,700	52	estimated, listing
Hoes, 2	1,898	36	estimated, listing
Lever ripper	1,451	28	estimated, listing
Wheelbarrows, 2	8,200	158	estimated, listing
Subtotal	15,749	303	

No broadfork was found listed in Ukraine; the lever ripper stands in for it until one is made or bought, [set].

6.7 Bays and store

Item	UAH	EUR	Basis
Compost bays, 1.0 m³ of board	12,900	248	estimated, listing
Metal tool store	16,900	325	estimated, listing
Subtotal	29,800	573	

6.8 Summary

Section	UAH	EUR
Ground, priced lines	48,049	924
Irrigation, priced lines	18,854	363
Tunnel	104,500	2,010

Section	UAH	EUR
Fence	130,648	2,512
Covers, priced lines	19,503	375
Tools	15,749	303
Bays and store	29,800	573
Construction subtotal	367,103	7,060
Design check, 8 percent	29,400	565
Contingency, 20 percent	73,400	1,412
Total external cost, priced lines	469,903	9,037

The manure line is the compost module's: 700 UAH a tonne for rotted cattle manure, Kyiv region, seen 2026-09-28, at 900 kg per m³; no bulk listing was found in western Ukraine, and a local, delivered quotation replaces it. The total leaves out the manure's carriage, the seed, the hoops, the tarps, the pond, its intake, pump, and fence, which are not priced. It is not the garden's cost until they are. The percentages are the working papers' own. For comparison only, Fortier lists start-up costs of 39,000 Canadian dollars for a garden of under two acres, including a heated greenhouse, two hoopouses, a two-wheel tractor, and a cold room (p. 9); none of those prices is used here.

Contribution, not in the total. None counted. The grower's work is a cost, not a contribution.

Running cost. The grower's pay, set each season by the association's pledges; its floor is the legal minimum wage, 8,647 UAH a month in 2026, for the months the garden is worked, [set]. Fortier writes that a full acre of diverse vegetables is a lot of work for one person, and that his 1.5 acres take two experienced growers full time with one full-time and one part-time worker (pp. 20–21); about three quarters of an acre here means one grower and seasonal help, [set], both paid. Seed, [set]. Tunnel film, rated for six seasons: about 22 m of 12 m film at a listed 640 UAH a metre, 14,080 UAH, length [assumed], about 2,350 UAH a season. Row cover, about three seasons in Fortier's use (p. 122). Soil analysis every few years; lime once in six to eight years. Drip tape replacement, [from L2]. Bought compost, about 11 to 21 m³ a year, 7,056 to 13,356 UAH at the same listing, carriage not included, from the compost module — the largest recurring line the garden's ground has.

7. Operation and maintenance

Responsible: the grower, with a deputy — [from L2]. One named role, paid, and seasonal help, paid; not a rota of residents.

- **Daily, in season:** [from L2]
- **Before each bed:** crop residues cut and covered with a black tarp for two to three weeks; broadfork or ripper; compost where the rotation gives it; raked level — Fortier's sequence (p. 42).
- **Seasonal:** covers on before a forecast frost; the tunnel closed at night in spring and autumn; spring crops under row cover.
- **Annual:** the crop plan written in winter, plot by plot, from the kitchen's menu; the bed record — what was sown, when, and what each bed gave — kept and carried into the next plan and into the bed_yield parameter, as Fortier keeps his (p. 139); the film checked; the fence walked.

If the programme works in the garden, that is described in the programme modules, in standing format. No produce, photograph, or share is tied to a child.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Stream low, driest month	[from L2]	fewer beds; the pond's reserve
Pump intake clogged	[from L2]	the sediment filter cleaned
Frost on open beds	[from L2]	covers, the tunnel; [from L2]
Compost short of the beds	[from L2]	fewer beds, then more
Acid-sensitive crops fail	[from L2]	lime to the analysis
Animals through the fence	[from L2]	[from L2]
One grower, no deputy	[from L2]	the association's terms
Produce refused	[from L2]	registration before the harvest

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-09-28, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Instruments found on 2026-09-28, read on secondary copies of the statute; each is read on zakon.rada.gov.ua and signed before this row counts as checked. Law No. 771/97-BP of 23.12.1997 on food safety and quality, as amended to Law No. 4718-IX of 16.12.2025: primary production of plant products needs no operating permit (Art. 23) but its capacity is registered, free, at least ten days before starting (Art. 25; Minagro Order No. 431 of 15.02.2024); HACCP does not apply to primary production (Art. 21); home-made food may be sold only at agri-food markets (Art. 37), where plant products are tested by the market's laboratory (Art. 36). Cabinet Resolution No. 305 of 24.03.2021: a children's establishment is supplied by a food market operator, with documents. MOH Order No. 377 of 20.03.2026: the sanitary regulation for children's health and recreation establishments of any ownership, catering included. Law No. 742-IV of 15.05.2003: a personal peasant household may sell its surplus, and that is not business activity. Tax Code, subparagraph 165.1.24: income from selling one's own produce is untaxed up to 2 ha, with a council certificate. Law No. 2496-VIII of 10.07.2018: no produce is called organic, bio, or eco without certification. Which of these binds this garden depends on its legal form, an open decision. The written enquiry to the district food-safety service, with the design drawn, comes before the first harvest reaches the kitchen. Taking water from the stream into a pond for irrigation may need a water-use permit, one of the written enquiries the compendium already names for the first year (Section 18); the pond's distance from the spring and its protection zone are asked in the same enquiry. Water Code, No. 213/95-BP of 06.06.1995, Arts. 88–89: a coastal protective strip of 25 m along small streams and ponds under 3 ha, doubled where the bank slopes more than 3°, in which vegetable growing, storing or applying fertiliser, and manure stores are not allowed; found 2026-09-28, read on secondary copies, not checked. Whether the strip applies around the garden's own

pond, and whether the pond may be dug inside the stream's strip, are asked in the same enquiry. A new law on organic production, No. 4921-IX of 30.06.2026, most of it in force from 2029, is noted and not yet read.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A drier valley:** the water decides the garden's size before the ground does; size the source first, then the beds.
 - **A colder valley:** fewer open beds and more under cover; the tunnel and caterpillar tunnels first, and crops chosen to the measured season.
 - **A smaller group or less ground:** fewer plots of the same shape; keep sixteen beds to a plot and the rotation in pairs; the fence and tunnel do not shrink in proportion, so the cost per bed rises.
 - **A larger group or more ground:** more plots; past about an acre, Fortier's advice is a second grower, not a machine (pp. 20–21).
 - **Neutral soil:** leave out the lime; keep the analysis.
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12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
- **From whose knowledge:** Jean-Martin Fortier, *The Market Gardener: A Successful Grower's Handbook for Small-Scale Organic Farming*, New Society Publishers, 2014, for the bed and plot system, the rotation, the compost programme, the water figure, the fence, the covers, the labour, and the crop-planning method, cited by page; Charles Dowding, for the no-dig compost depths; John Jeavons and Ecology Action, and Christopher Alexander's *Vegetable Garden*, for the comparison of areas; the rational norm of vegetable consumption and Cabinet Resolution No. 305 for demand; a Ukrainian agricultural guide of 2025 for yields; a retailer's guide and an agronomic table for liming; the compost module for the compost bought in; Ukrainian retail and service listings of 2026-09-28 for prices.
- **Traditional knowledge:** none. Where vegetables were grown, how the ground was fed, what was grown, and whose seed, are for the elders.
- **Text:** CC BY-SA 4.0
- **Drawings and designs:** [hardware licence — open decision]
- **Measurement data:** none yet
- **Gate 1:** household decision, 2026-09-29, recorded by Michel Garand — v0.5 may be published as it stands, as a marked draft, in English, Ukrainian, and German

- **Wartime review:** 2026-09-29, Michel Garand — v0.5 read against compendium Section 2; nothing places the site; the note on permits and local rules added; accepted
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13. What to check

- **The stream and the pond** — the stream's flow in its driest month, by bucket and stopwatch as the water module measures the spring; at least about 5 L a minute carries the full garden; the pond's volume set from it; the water-use permit; the pump, its power, and its circuit with the energy module. Until the flow is measured, the garden is sized by its water, not by its ground.
 - **Establishment compost** — a local, delivered quotation for about 62 m³ of rotted manure or compost, replacing the Kyiv-region listing, and its carriage.
 - **The protective strip** — the stream's low-water line and the pond's edge marked on the walk, the bank slopes measured to settle 25 or 50 m; the beds, bays, and heap placed outside it; the enquiry on the pond.
 - **Seed, hoops, and tarps** — priced from a crop plan and the bed length.
 - **The season** — frost_free_days from the nearest station, then the site's own record.
 - **Altitude and climate** ranges for Section 2.
 - **The soil** — the analysis, replacing the assumed loam and pH and the dolomite rate.
 - **The ground** — the walk: whether about 3,000 m² of ground under 5 percent, facing south to south-east and out of the frost pocket, exists without taking the best hayfield; the contour length that sets the bed; replaces the 220 m of fence and the 120 m of main line.
 - **vegetables_per_person** — the kitchen's own menu, replacing the rational norm.
 - **bed_yield** — the garden's own record, bed by bed, from the first season; Fortier's yields per bed (p. 141) scaled to 20 m beds for comparison.
 - **garden_share** — set by the association and the kitchen.
 - **Resident days of demand** — once resident households are known.
 - **The grower and seasonal help** — months, hours, and pay, set with the association.
 - **Resolution No. 305** — the portion tables read on zakon.rada.gov.ua, confirming our reading of 0.40 to 0.46 kg a day.
 - **The Ukrainian row** — every instrument read and signed; the written enquiry to the district food-safety service.
 - **Quotations** for the lines in Section 6, dated, to replace the listings.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the grower's daily tasks in Section 7; early signs and repair costs in Section 8; Section 9.
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